Field Hours

CSCE 420 Artificial Intelligence

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 50 minutes

An agent and a search

An agent takes a percept and returns an action. A problem, for search, is a state, a set of actions, a step cost, and a goal test. The solution is a sequence of actions, not a single clever number. Uninformed search tries actions without a hint about which one is closer to the goal.

Breadth-first search expands the shallowest open state. On a tree with a uniform step cost it finds the fewest actions. It also stores every state on the frontier. Depth-first stores a path, not a layer, and can miss a short solution if it dives. Name the cost you care about before you pick the search.

Breadth-first, one layer.

1. Expand the start

2. Enqueue its children

3. The first time you reach the goal, the path is shortest in hops

A corridor of rooms

1. Rooms 1-2-3-4 in a line, and a side door from 2 to the goal. Start at 1. Each move costs 1.

2. Breadth-first reaches the goal through 2. The path has two moves. Depth-first might walk 1-2-3-4 before it backs up, if it tries the corridor first.

3. The goal test runs when you take a state off the frontier to expand it, not when you first enqueue it, if you want the shortest path. Checking at enqueue is a shortcut that is safe only for uniform cost.

Problem 1. The step costs are not equal. Is breadth-first still the fewest-cost path?

Answer. No. Breadth-first counts actions, not cost. Use a search that expands the lowest path cost so far. A cheap long path can beat an expensive short one.

Problem 2. Why does CSCE 633 accept this course or CSCE 625?

Answer. Both teach search and a first probabilistic model. If 625 is already one of your ten master's courses, you can skip 420. Do not skip both and open machine learning cold.

Lecture 2 · 50 minutes

A heuristic

A heuristic estimates the remaining cost from a state to the goal. It is not the true remaining cost. A* expands the state with the smallest path-so-far plus estimate. If the estimate never exceeds the true remaining cost, the first

time A* expands the goal the path is optimal. That property is called admissible.

An estimate that sometimes overshoots can return a path that is not cheapest. An estimate of zero is admissible and useless: A* becomes uniform-cost search. A good heuristic is admissible and large. Larger means fewer expansions, as long as you did not lie upward.

Two estimates of the same drive.

Admissible

Straight-line miles, which a road cannot beat

A* may trust it

Not admissible

A guess larger than the real road

A* may skip the true best path

Three towns

1. You have driven 10 miles to town B. The straight line from B to the goal is 6. The score is 16.

2. The other open town, C, cost 4 miles to reach and the straight line is 14. The score is 18. Expand B first.

3. If someone had claimed the line from B was 20, and the real road was 6, the algorithm might expand C and stop on a worse path. The lie was the overestimate.

Problem 1. Your heuristic is twice the straight-line distance. Is A* still optimal?

Answer. Not in general. Twice a true lower bound can exceed the true remaining cost. Optimality needs the estimate to be at most the truth.

Problem 2. The heuristic is perfect: it equals the true remaining cost. How many states does A* expand, beyond the solution path?

Answer. Essentially none. Every detour has a higher score than the true path. A perfect heuristic is rare. It is the upper end of what admissible means, not a typical map.

Lecture 3 · 50 minutes

Games

A two-player game of perfect information is a tree. You move, then the opponent, then you. Minimax scores a leaf and assumes the opponent minimizes your score while you maximize it. The value of a position is the best you can force, not the best leaf that exists if the opponent plays badly.

The whole tree is too large. Alpha-beta prunes a branch once it cannot change the decision already available. The answer stays the same as minimax if both sides play the same evaluation. The order of moves matters for speed, not for the value, when the evaluation is exact.

One ply, then the reply.

Your three moves

The opponent's replies under each

You keep the max of the mins

Two moves, two replies

1. Move A has replies worth 3 and 1. The opponent picks 1.

2. Move B has replies worth 2 and 4. The opponent picks 2.

3. You pick B, value 2. The leaf worth 4 is not available, because the opponent will not hand it to you. A program that returns 4 is assuming a friendly opponent.

Problem 1. You stop the search at a fixed depth and score the position with a guess. What did you give up?

Answer. The exact value of the game. The score is only as good as the guess and the depth. A horizon just before a loss can look fine. That is why the evaluation and the depth are part of the claim, not an implementation detail.

Problem 2. Alpha-beta cuts a branch. Can the chosen move differ from minimax?

Answer. No, if the leaf scores are the same and both algorithms finish. The cut only skips a branch that cannot beat the best alternative already known. A bug in the bounds can change the move. The idea does not.

Lecture 4 · 50 minutes

Constraints

A constraint problem is variables, domains, and constraints that rule out some combinations. The solution assigns every variable a value that satisfies every constraint. Backtracking tries a value and retreats when a constraint is already broken. Forward checking, after an assignment, deletes values in other domains that the new assignment forbids.

The order of variables changes the size of the tree. A variable with two legal values left is a smaller branch than a variable with twenty. Pick the tight one first. This is a heuristic, not a proof that the problem is easy.

One assignment.

1. Pick an unassigned variable

2. Try a value still in its domain

3. Delete values that now conflict

4. Backtrack if a domain goes empty

Three colors, three regions

1. Regions X, Y, Z. Colors red, green, blue. X touches Y. Y touches Z. X does not touch Z.

2. Set X to red. Forward checking removes red from Y. Z still has all three colors.

3. Set Y to green. Z may still be red, because X and Z do not touch. Blue and green remain too, except green was taken by Y if they touch. Z ends with red and blue. Two solutions from here, not a failure.

Problem 1. A domain becomes empty. What do you do, and what do you not conclude about the whole problem?

Answer. Undo the latest assignment and try another value. An empty domain kills that branch, not the problem. Another color earlier might have worked.

Problem 2. Why is 'the map has a solution' not the same as 'my search found one'?

Answer. The search might have stopped early, or a constraint might be coded wrong. A found assignment you can check against every constraint is evidence. A failed search is evidence only if the search was complete.

Lecture 5 · 50 minutes

One probabilistic model

A probability is a number between 0 and 1 for a proposition you are not treating as certain. The useful step is conditioning: the chance of A given B is the chance of both, divided by the chance of B, when B is possible. Bayes' rule swaps the condition when the swapped one is the quantity you can estimate.

A Bayes net is a picture of which variables depend on which. An arrow is a claim of direct dependence, not a decoration. If the picture says a symptom depends on a disease and not on an unrelated test, you do not invent an edge because the numbers moved once. CSCE 633 will fit models like this. Here you only compute one update by hand.

A single update.

Prior on the cause -> How likely the observation is if the cause is true -> How likely the observation is if it is false -> Posterior

A rare fault and a test

1. One machine in a hundred has the fault. The test catches the fault 90 times out of 100, and it false-alarms 5 times out of 100 on a healthy machine.

2. Out of 10,000 machines, about 100 are faulty and 90 of those alarm. About 9,900 are healthy and about 495 of those alarm.

3. An alarm is therefore 90 true out of 585 alarms, about 15 percent. The test is accurate and the fault is still uncommon among the alarms. Ignore the base rate and you will quote 90 percent, which is the wrong question.

Problem 1. The test is positive. Which number is not the answer: the true-positive rate, or the chance of the fault given the alarm?

Answer. The true-positive rate is 90 percent. The chance of the fault given the alarm is about 15 percent in this story. They are different conditional probabilities. Bayes' rule is the swap.

Problem 2. You add an arrow from the weather to the fault because one rainy day had more faults.

What is wrong with that as a model?

Answer. One day is not a dependence you have earned. An arrow is a claim you will use for every later inference. Add it when the mechanism or the data supports it, and then the joint distribution changes. A story is not an edge.

Lecture 6 · 50 minutes

What learning will assume

You can now separate a search from a heuristic, a game value from a friendly leaf, a constraint failure from an empty search, and a true-positive rate from a posterior. CSCE 633 starts at fitting a predictor, not at the definition of a state.

If CSCE 625 is on your master's plan, that course covers this ground at graduate depth and the catalog lets it stand in for 420. Taking both 420 and 625 is allowed as study. Only 625 sits inside the 36 credits.

Four sentences to keep.

A solution is a path with a cost

An admissible heuristic does not overestimate

The opponent will not choose your favorite leaf

An alarm is not the base rate

A question you should answer cold

1. A test is 99 percent accurate and the condition is rare. Why might most alarms be false?

2. Because the healthy population is large. A small false-alarm rate times a large population can exceed the true alarms.

3. If that sentence is not immediate, redo the numerical update before CSCE 633.

Problem 1. Name the catalog line that lets you skip this course.

Answer. CSCE 633 requires CSCE 420 or CSCE 625. Skipping 420 is legitimate when 625 is one of the ten. Skipping both is not.

Problem 2. A* with an admissible heuristic expands the goal. What may you claim?

Answer. The path cost is optimal. You may not claim it expanded the fewest nodes among all possible heuristics. You may not claim the heuristic was perfect.